Extra Sheet : Discussion

EGCI341: SOFTWARE ENGINEERING (WEEK 01)

Code of Ethics – Eight Principles

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Public:

- Software engineers shall act consistently with the public interest.

Client and Employer:

- Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

Product:

- Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

Code of Ethics – Eight Principles

Judgment:

- Software engineers shall maintain integrity and independence in their professional judgment.

Management:

- Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

Code of Ethics – Eight Principles

Profession:

- Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

Colleagues:

- Software engineers shall be fair to and supportive of their colleagues.

Self:

- Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Public

Software engineers shall act consistently with the public interest.

Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment, that they reasonably believe to be associated with software or related documents.

Client & Employer

Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

Promote no interest adverse to their employer or client, unless a higher ethical concern is being compromised; in that case, inform the employer or another appropriate authority of the ethical concern.

Product:

Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

Judgment:

Software engineers shall maintain integrity and independence in their professional judgment.

Management

Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

"To be a good manager, you must confuse your staff. If they can't understand you, they'll think that you're brilliant."

Ensure good management for any project on which they work, including effective procedures for promotion of quality and reduction of risk.

Profession

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

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Be accurate in stating the characteristics of software on which they work, avoiding not only false claims but also claims that might reasonably be speculative, vacuous, deceptive, misleading, or doubtful.

Colleagues

Colleagues:

Software engineers shall be fair to and supportive of their colleagues.

Assist colleagues in professional development.

Self

Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Further their knowledge of developments in the analysis, specification, design, development, maintenance, and testing of software and related documents, together with the management of the development process.

Section 1:
Ethical Dilemmas in Professional Software Development

Scenario 1.1

- You are asked to develop software that stores and manages customers financial data.
- You find a security flaw days before its delivery to the client.
- Your boss tells you to sit on it and will be fixed with a patch after the delivery.
- If you bring the flaw to the foreground it will delay the release and cost your company millions.
- What do you do?

Scenario 1.2

- You work for a small software firm that is contracted to write a program that can predict the spread of radiation of a dirty bomb with 75% accuracy.
- The best you have been able to achieve is 74.6% accuracy.
- Your boss says close enough and rounds up the test data.
- What do you do?

Ethical Dilemmas

- Disagreement in principle with policies of senior management
- Your employer acts unethically by releasing a safety-critical system without finishing the testing of the system
- You participate in the development of military weapons or nuclear systems

Section 2:
AI Ethical Dilemmas in Professional Software Development

Scenario 2.1: The "High-Stakes" Recruitment Filter

The Situation: You are a Lead Developer at a major FinTech firm. Your manager tasks you with deploying "EliteScreen," an AI tool designed to filter through 50,000 resumes for 10 open positions.

During testing, you notice the AI consistently assigns lower scores to candidates who attended "Historically Black Colleges and Universities" (HBCUs) or who have gaps in their resumes (often associated with parental leave).

Your manager insists on deploying it immediately to meet hiring targets, arguing that the AI is simply "learning from the successful profiles of our current top performers."

What do you do?

Scenario 2.2: The "Nudge" for Productivity

The Situation: Your startup is building a project management tool. The CEO wants to add a "Predictive Productivity" feature that uses AI to monitor employee keystrokes, webcam "sentiment analysis" (to see if they look tired or bored), and active window time.

The goal is to "alert managers when a developer is at risk of burnout."

However, you discover the backend is actually designed to generate "Low Performer" reports used for secret layoffs.

What do you do?

Scenario 2.3: The "Black Box" Medical Triage

The Situation: You are a Software Architect for a hospital system. You’ve developed a "Neural Triage" AI that predicts which patients in the Emergency Room need immediate surgery.

The AI is 15% more accurate than human doctors. However, it is a "Black Box" model—you cannot explain why it chooses Patient A over Patient B. A family sues the hospital because their relative was de-prioritized by the AI and suffered a stroke.

The legal team asks you to "adjust the logs" to make it look like a doctor made the final call.

What do you do?

Scenario 2.3: The "Priority Route" Algorithm

The Situation: You are a senior engineer for a Smart City project developing an AI-driven traffic management system. The goal is to minimize overall commute times.

Your analysis shows that the AI has achieved a 20% reduction in city-wide traffic.

However, you discover the AI achieved this by rerouting heavy congestion through lower-income residential neighborhoods with older infrastructure and fewer noise barriers, while keeping "Executive Corridors" and high-income districts clear. Your stakeholders are thrilled with the 20% "success" metric.

What do you do?

Scenario 2.5: The "Shadow Training" Repository

The Situation: Your team is under a "crunch" period to deliver a complex cybersecurity module. A junior dev finds a specialized Generative AI tool that writes perfect, bug-free code for this specific niche.

You later discover that this AI was trained on a massive leak of private, proprietary code from a direct competitor.

Using the AI's output technically means your company's product will contain "stolen" logic patterns, but the code itself is functionally unique and impossible to trace.

What do you do?

Scenario 2.6: The "False Positive" Safety Override

The Situation: You are building a DevOps pipeline for an autonomous rail system. You’ve implemented an AI-driven "Predictive Maintenance" bot that can automatically halt trains if it detects a 90% probability of a mechanical failure.

During a peak holiday weekend, the AI triggers a "Halt" on the main line. Your lead sensor engineer insists it’s a "false positive" based on a faulty sensor.

If you override the AI, the trains keep moving and the company avoids millions in refunds. If the AI is right and you override it, a derailment is possible.

What do you do?

Note:

Don't just build things that work—build things that are *right*.

Reference

John Eveland, John Hoffstatter, “Software Engineering Code of Ethics”, http://n1.slideserve.com/PPTFiles/Talk-Software_Ethics_46833_49926.ppt